

Total No. of Questions : 8]

SEAT No. :

PE4255

[6582]-26

[Total No. of Pages : 2

**S.E. (Electronics /E &TC/Electronics (VLSI Design & Tech.)/E.C
(A.C.T.))**

ELECTRONIC CIRCUITS (EC)

(2019 Pattern) (Semester - III) (204181)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data, if necessary.
- 4) Neat diagram must be drawn wherever necessary.

- Q1)** a) Define Voltage regulation. Explain Concept of Low drop out Voltage regulator. **[6]**
- b) IC Voltage Regulator using IC 317, Calculate values of R2 for the output voltage 5v to 10v, assume $R_1=240\Omega$ and $I_{adj} = 100\mu A$. **[6]**
- c) Draw and explain with Block diagram Switch Mode Power Supply (SMPS). **[6]**

OR

- Q2)** a) Draw and explain the block diagram of Power Supply. **[6]**
- b) IC Voltage Regulator using IC 317, Calculate values of R2 for the output voltage 15v to 30 v, assume $R_1=240\Omega$ and $I_{adj} = 100\mu A$. **[6]**
- c) Write a Short note on “Current boosting Voltage regulator”. **[6]**
- Q3)** a) Draw Block diagram of OP-AMP. And explain the function of each block. **[6]**
- b) Define the following Characteristics of OP-AMP **[6]**
- i) Input offset Current
 - ii) Input offset voltage
 - iii) Slew Rate
- c) Compare Concept of open loop and Close loop configuration of OP-AMP. **[5]**

OR

- Q4)** a) State Ideal and Typical values of OP-AMP parameters (IC 741). **[6]**
- b) Draw the equivalent circuit of OP-AMP with ideal transfer curve. **[6]**
- c) Explain the need of level shifting in OP-AMP. **[5]**

P.T.O.

- Q5)** a) Draw and explain non-Inverting amplifier with its input and output waveforms. [6]
b) Design a Practical differentiator circuit for the input signal having maximum frequency of operation 150 Hz. [6]
c) Draw and explain Instrumentation amplifier circuit using three OP-AMP. [6]

OR

- Q6)** a) Compare Inverting and Non-Inverting amplifier in OP-AMP. [6]
b) Design a Practical Integrator circuit to operate at $f = 2$ KHz and gain equal to 4. [6]
c) Draw and explain Zero Crossing detector using OP-AMP. [6]
- Q7)** a) Draw and explain V to I Converter using grounded load using OP-AMP. [5]
b) Explain the disadvantages of binary weighted resistor circuit over R/2R DAC using OP-AMP. [6]
c) With the help of neat block diagram explain operation of PLL. [6]

OR

- Q8)** a) Draw and explain I to V Converter using OP-AMP. [5]
b) Draw and explain the circuit of flash type ADC using OP-AMP. [6]
c) Define the following specifications of ADC [6]
i) Resolution
ii) Accuracy
iii) Quantization errors

